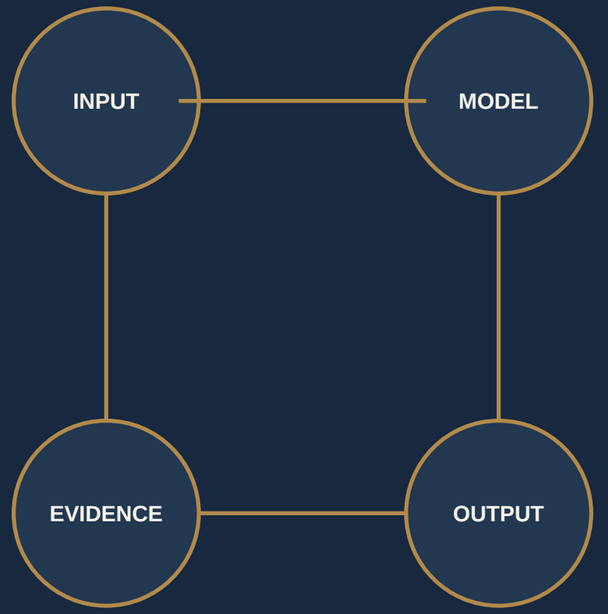
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Cyber-Physical Systems

AI.302 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define the integration of computation.
- Apply key methods including networking.
- Build or analyze a practical system involving sensors.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

The integration of computation

02

METHODS

Networking

03

SYSTEMS

Sensors

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · The integration of computatio

The integration of computation

02 · Networking

Networking

03 · Sensors

Sensors

04 · Actuators and physical process

Actuators and physical processes

05 · Mandatory practical: Design an

Mandatory practical: Design and simulate a cyber-physical robotic system

Core ideas, made explicit

AI.302 • Distinguish definitions, assumptions and evidence

- The integration of computation
- Networking
- Sensors
- Actuators and physical processes
- Mandatory practical: Design and simulate a cyber-physical robotic system

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.302 • Distinguish definitions, assumptions and evidence

- The integration of computation
- Networking
- Sensors
- Actuators and physical processes
- Mandatory practical: Design and simulate a cyber-physical robotic system

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
The integration of computation

INPUTS
Networking

METHOD
Sensors

OUTPUT
Actuators and physical processes

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving the integration of computation.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using the integration of computation.
- Implement or simulate a baseline using networking.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.